Project Proposal: Text-to-Image Generation with CLIP and Stable Diffusion

Generative AI project | Team: [team name] | Date: [date]

1. Objective

Build a text-to-image pipeline that turns natural-language prompts into 512×512 images, using a curated COCO image–caption subset to (a) establish a reproducible zero-shot baseline with Stable Diffusion v1.5 and (b) measure whether lightweight LoRA fine-tuning improves prompt adherence on COCO-style scenes.

2. Dataset

- **Source:** MS-COCO 2017 captions (official annotations, CC BY 4.0), 5 human captions per image across everyday objects and scenes.
- **Curated subset:** 2,000 image–caption pairs, seeded random sample, split 1,600 train / 200 val / 200 test by image so no image crosses splits.
- **Cleaning:** ASCII normalisation, whitespace and punctuation fixes, drop captions under 4 or over 40 words (CLIP caps at 77 tokens), remove duplicate captions, require shorter image side ≥ 256 px; square centre-crop to 512×512.
- **Why COCO over Flickr30k:** broader object and scene variety, direct download from official URLs, standard benchmark for CLIP-based evaluation. Flickr30k is people-centric and has stricter access terms.

3. Model and approach

- **Architecture:** Stable Diffusion v1.5 latent diffusion. A frozen CLIP ViT-L/14 text encoder produces 77×768 token embeddings; a UNet denoises in the VAE latent space with cross-attention to the text.
- **Baseline (Phase 1):** zero-shot SD v1.5, DPM-Solver++ with 30 steps, guidance scale 7.5, five fixed prompt–seed pairs so every later model is compared on identical inputs.
- **Adaptation (Phase 2):** LoRA on UNet attention layers (rank 8–16) trained on the 1,600 train pairs in fp16. Ablations: single vs. all five captions per image, and LoRA rank.
- **Stack and compute:** PyTorch, Hugging Face Diffusers, Transformers (CLIP); fits a 16 GB GPU (Colab T4); GitHub for version control.

4. Evaluation

- CLIPScore (prompt adherence) on the 5 test prompts and on the 200 held-out test captions; text→image retrieval recall on val as a data-quality check.
- FID/KID of generated vs. real test images (noted as noisy at n=200), plus a small blind human preference comparison of baseline vs. LoRA outputs.
- **Deliverables:** this proposal, 5 sample prompt→image test runs, and an initialised GitHub repo with data pipeline, baseline notebook and tests.

5. Team roles

Role	Owner	Responsibilities
Data lead	[name]	Dataset curation, cleaning and preprocessing pipeline; split and dataset statistics.
Modeling lead	[name]	Environment, SD v1.5 baseline, LoRA fine-tuning runs, hyperparameter ablations.
Evaluation & integration lead	[name]	CLIPScore, retrieval and FID/KID metrics, fixed-seed prompt panel, human preference check; GitHub repo, tests, README, final report.